

Robotics, Artificial Intelligence and Embedded Systems



**Cognitive Systems**

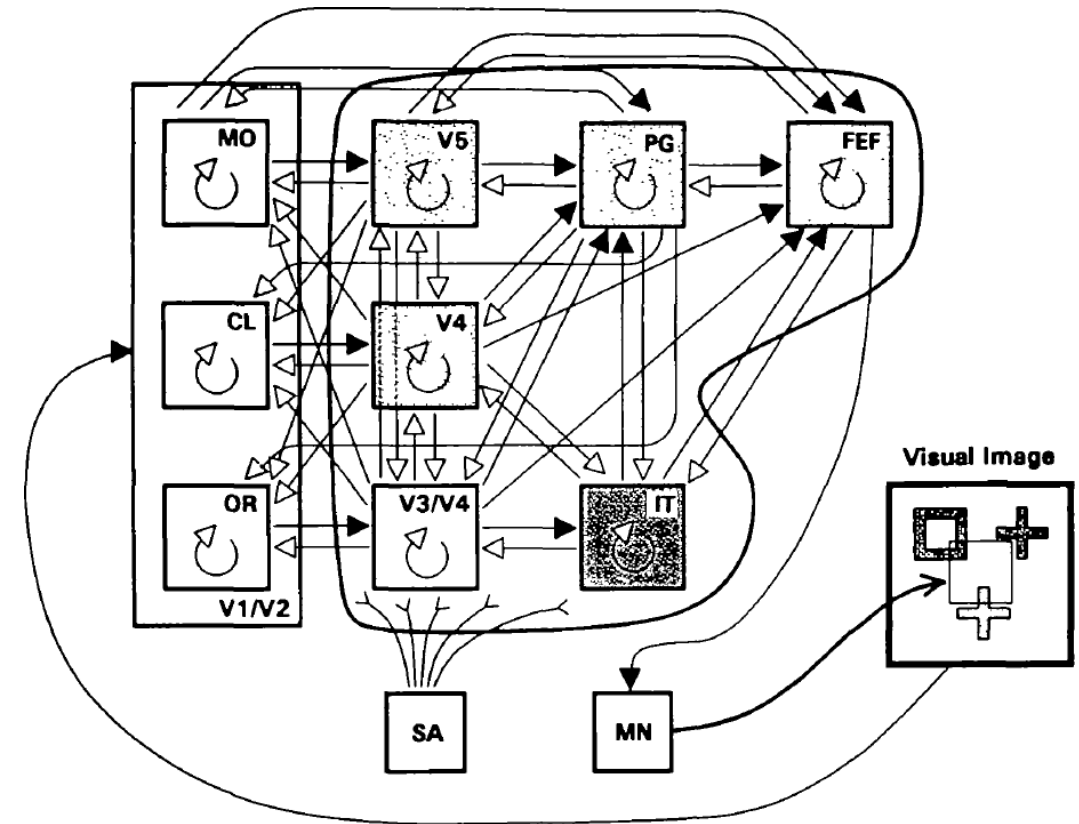
# Neurons and the Brain III

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# Example: Visual Cortex (1992)

One of the first large-scale brain simulations was a **task-tailored engineered** computer simulation of the visual cortex with nine functionally segregated areas by Tononi et al. (1992).

- For efficiency, neurons are not simulated individually but as **neuronal groups**
- The model...
  - ... is comprised of **10,000** neuronal groups and **1,000,000** connections
  - ... processes input from a **color camera**
  - ... solves the **binding problem** (link object attributes coherently across the different cortical areas)

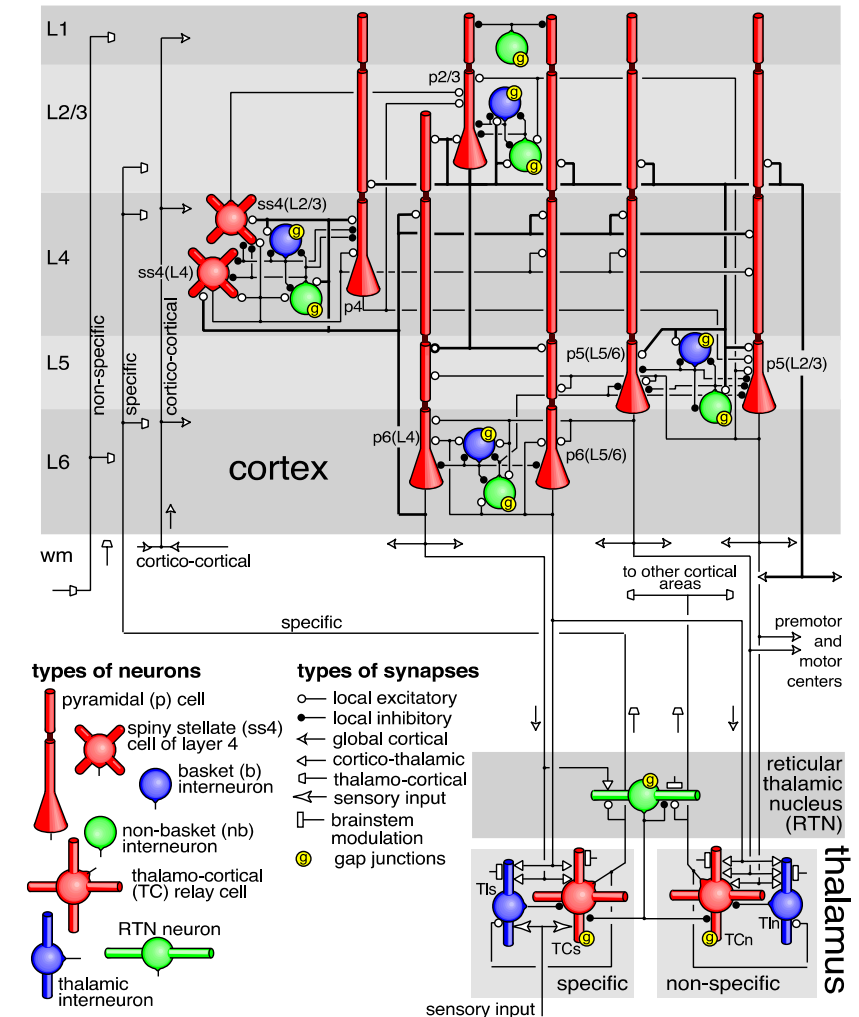


From Tononi, G., Sporns, O., Edelman, G.M., 1992. Reentry and the Problem of Integrating Multiple Cortical Areas: Simulation of Dynamic Integration in the Visual System. *Cereb Cortex* 2 (4), 310–335.

# Example: Thalamocortical System (2008)

The **engineered** thalamocortical brain model by Izhikevich et al. (2008) is based on DTI measurements and 3D neuron reconstructions.

- The model contains **one Million** spiking multi-compartment neurons and almost **500 Million** synapses
- The simulation of the neural dynamics is based on the **Izhikevich** neuron model; there are 22 different neuron types for the thalamus and the different layers of the cortical microcircuits
- The model is able to **reproduce** typical patterns of brain activity

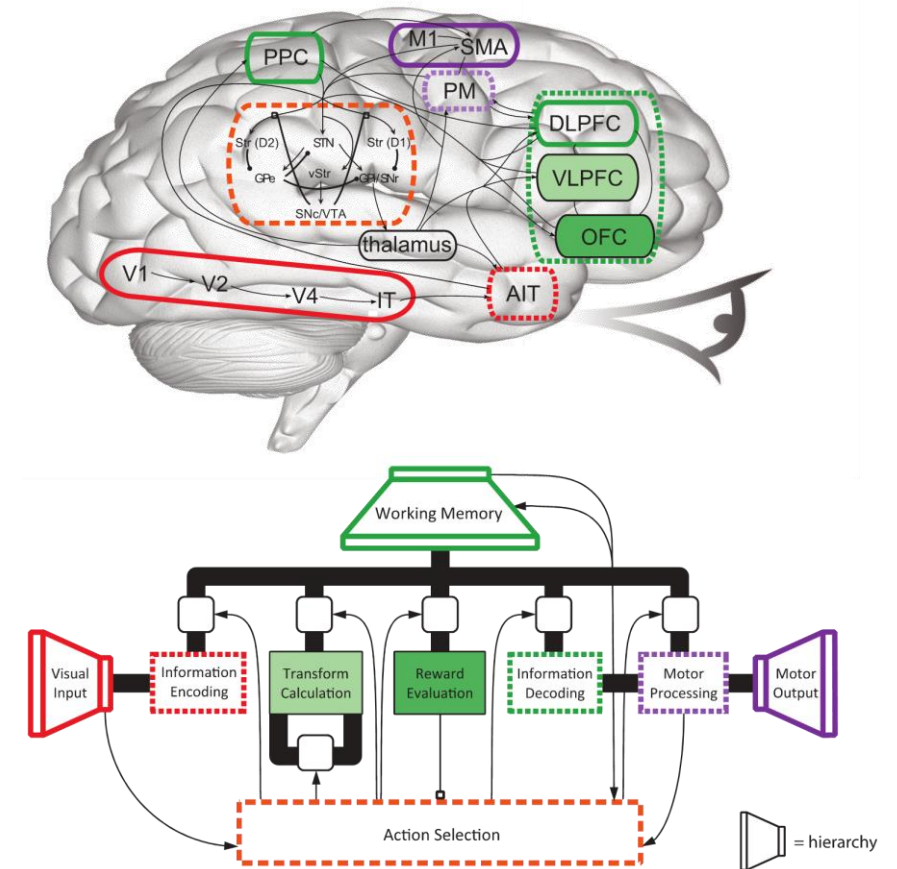


From Izhikevich, E.M., Edelman, G.M., 2008. Large-scale model of mammalian thalamocortical systems. *Proceedings of the National Academy of Sciences* 105 (9), 3593–3598.

# Example: Spaun (2012)

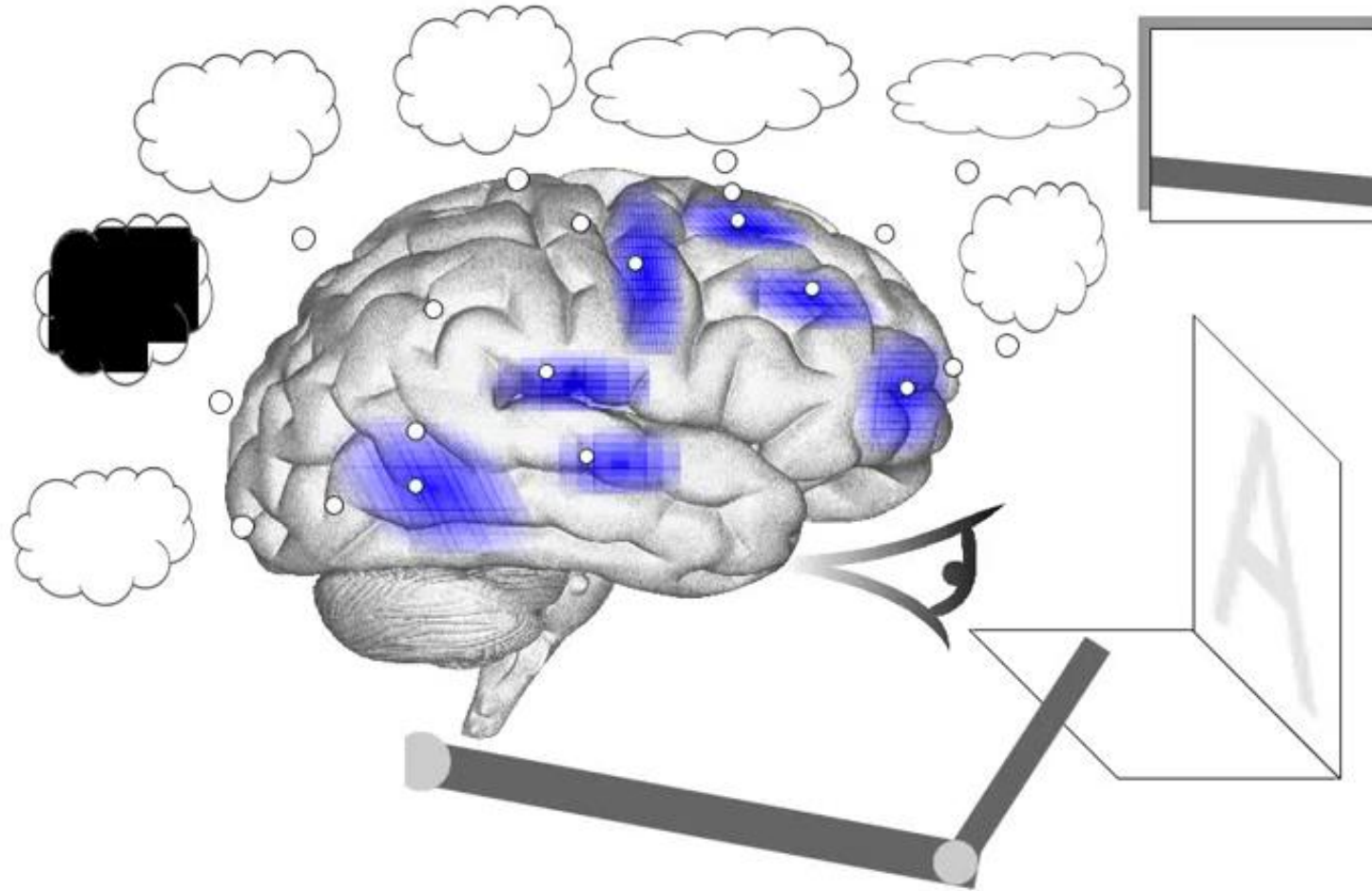
Spaun is a **functional** spiking neuron-based whole brain model that was **synthesized** from an abstract **cognitive architecture** based on the **Neural Engineering Framework** and the **Semantic Pointer Architecture**.

- The model is built from **2.5 Million** spiking neurons
- The cognitive architecture underlying the model implements **cognitive functions** such as image recognition, working memory, perception and action
- The focus of the model is on the **production of behavior**; it is therefore constrained by neuroscientific data but not derived or abstracted from it



Eliasmith, Chris; Stewart, Terrence C.; Choo, Xuan; Bekolay, Trevor; DeWolf, Travis; Tang, Yichuan; Rasmussen, Daniel (2012): A Large-Scale Model of the Functioning Brain. In Science 338 (6111), p. 1202. DOI: 10.1126/science.1225266.

# Demo: Sequence Learning (and Forgetting)



From <https://www.youtube.com/watch?v=XxlzmkWygIY>

# Intermediate Summary

All brain models considered so far were built based on a **specific set of assumptions** and for a **specific purpose**.

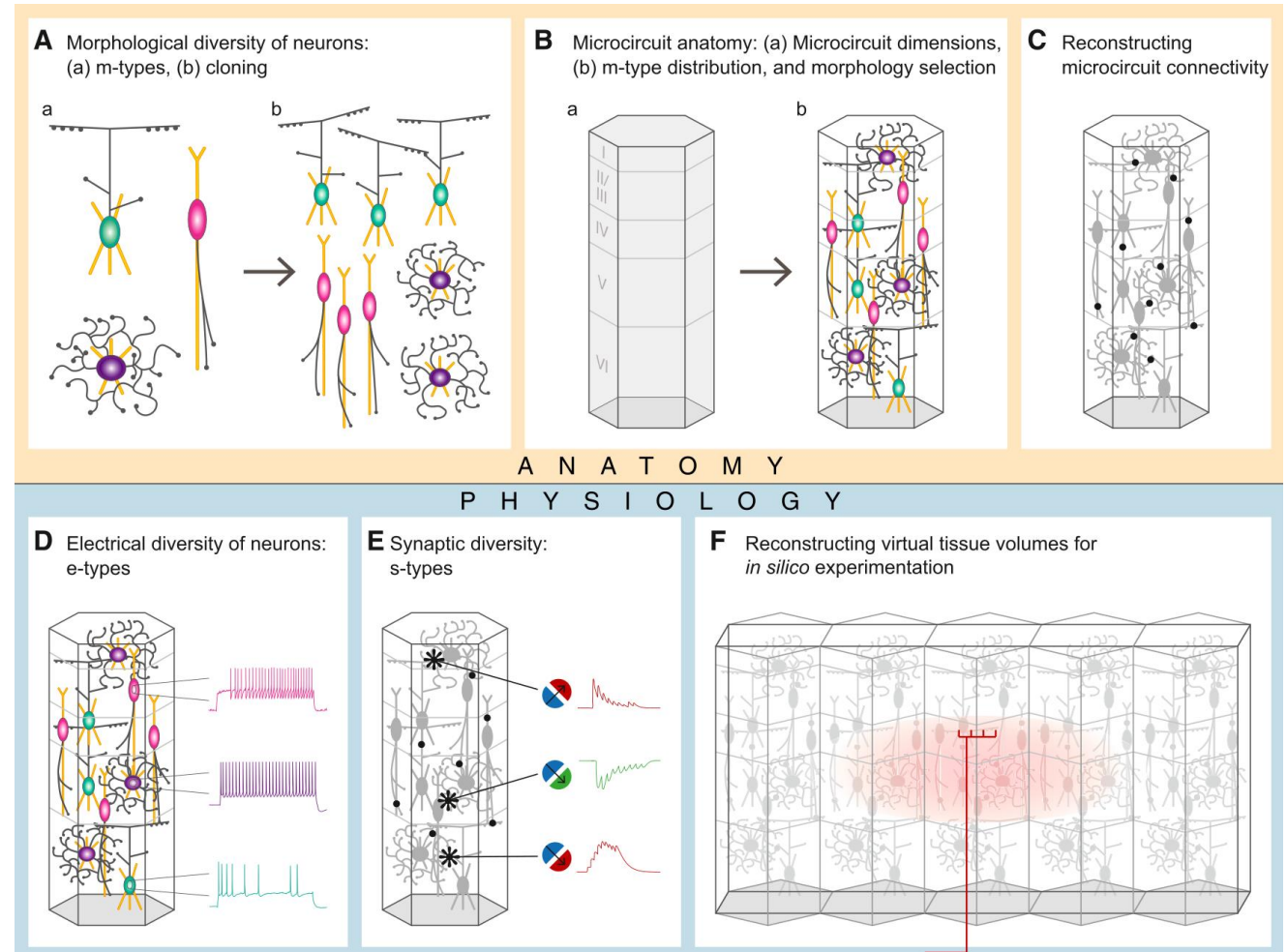
- All models are largely **decoupled** and correspond to different directions of research even though they represent the same ground truth – the brain
  - The manual engineering applied to derive these models makes it **hard to incorporate new findings** or to extend the them to represent additional aspects
- There is a clear need for a **flexible unified model** that captures **all knowledge** about the brain available to date and which can be easily **extended** to accommodate new data



# The Blue Brain Project (2015)

The Blue Brain Project has developed a **data-driven reconstruction process** for cortical microcircuits.

- The microcircuit consists of **31,000** anatomically detailed neurons and **37 Million synapses**
- The microcircuit model is a **digital reconstruction** that is based on both experimental **data** and **predictions**
- The simulation **reproduces data** recorded in experiments



From Markram, H., Muller, E., Ramaswamy, S., Reimann, M.W., et al., 2015. Reconstruction and Simulation of Neocortical Microcircuitry. Cell 163 (2), 456–492.